

Thermally developing flow and heat transfer in rectangular microchannels of different aspect ratios

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Abstract

Laminar convective heat transfer in the entrance region of microchannels of rectangular cross-section is investigated under circumferentially uniform wall temperature and axially uniform wall heat flux thermal boundary conditions. Three-dimensional numerical simulations were performed for laminar thermally developing flow in microchannels of different aspect ratios. Based on the temperature and heat flux distributions obtained, both the local and average Nusselt numbers are presented graphically as a function of the dimensionless axial distance and channel aspect ratio. Generalized correlations, useful for the design and optimization of microchannel heat sinks and other microfluidic devices, are proposed for predicting Nusselt numbers. The proposed correlations are compared with other conventional correlations and with available experimental data, and show very good agreement.

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1. Introduction

The increased power dissipation and shrinking dimensions of microelectronics devices have accentuated the need for highly effective compact cooling technologies. Microchannel heat sinks are of particular interest due to the very high rates of heat transfer they enable in conjunction with greatly reduced heat sink length scales and coolant mass. A significant amount of research has been dedicated to microchannel transport in recent years, as reviewed in [1,2]. Although, a number of published results on heat transfer in microchannels have differed from the behavior expected at “conventional,” i.e., larger, length scales, recent work by Lee et al. [3] found that numerical predictions based on a classical, continuum approach are in good agreement with the experimental data when the entrance and boundary conditions imposed in the experiment are carefully matched in the simulation.

The unique characteristics of microchannel heat sinks (small length scales, conductive substrate, abrupt contraction/expansion at the entrance/exit, and high pressure drop) give rise to conditions that are quite different from those of conventional channels. For abrupt contraction/expansion at the entrance/exit, Rohsenow et al. [4] suggested that the inlet condition should be assumed as being hydrodynamically fully developed but thermally developing, due to wake effects at the abrupt entrance prior to the channel. Lee et al. [3] verified this using a computational analysis which included inlet and exit manifolds and concluded that flows in microchannel heat sinks typically experience thermally developing laminar flow, with significant entrance effects. It is important for the design of microchannel heat sinks to be able to accurately predict the heat transfer coefficients under different flow and thermal conditions, especially in developing laminar flow. In contrast to ducts with cross-sections defined by a single coordinate, such as circular tubes and parallel plates, where analytical solutions may be readily obtained, the analysis of heat transfer in a rectangular duct is more complicated

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(CFD) computations can be performed with high accuracy and flexibility.

In the present work, laminar flow and heat transfer in the thermal entrance region of rectangular ducts with aspect ratios ranging from 1 to 10 is investigated for thermally developing flow with the H1 thermal boundary condition. A finite volume approach is employed to obtain the temperature and heat flux distributions on the channel walls. The local and average Nusselt numbers in the entrance region are numerically calculated as functions of the dimensionless axial distance and channel aspect ratio. Generalized correlations for both the local and average Nusselt numbers in the thermal entrance region are proposed. The results are compared to predictions from correlations for conventional channels as well as with experimental data for microchannels, and good agreement is noted. The proposed correlations are easy to use, provide detailed heat transfer coefficient predictions in the entrance region of microchannels, and cover a wide parameter range.

2. Mathematical formulation

The following assumptions are made to model the heat transfer in the rectangular channel:

- (1) steady state,
- (2) incompressible fluid,
- (3) laminar flow,
- (4) constant fluid properties,
- (5) negligible axial conduction and viscous dissipation, and
- (6) negligible radiative and natural convective heat transfer from the microchannel heat sink.

The dimensions of the microchannels considered in this work are listed in Table 1. To achieve the H1 boundary condition, a very thin and highly conductive wall (but with no axial conduction) is included in the model. The width (a) and axial length (l) of the channel in the computational model are held constant at 200 μm and 120 μm , respectively, while the height (b) of the channel is varied. The

aspect ratio of the rectangular channel and its hydraulic diameter are defined as

$$\alpha = \frac{b}{a} \quad \text{and} \quad D_h = \frac{2ab}{a+b} \quad (1)$$

Fig. 1 shows a schematic diagram of the microchannel cross-section considered. Only a quarter of the microchannel was included in the computational domain, in view of the symmetry conditions.

The governing equations for mass, momentum and energy are solved with boundary conditions as follows. The velocity is zero on all wall boundaries, and as the flow is assumed to be hydrodynamically fully developed, the following exact analytical solution by Marco and Han [10] is used as the fully developed inlet velocity profile:

$$\left. \begin{aligned} u(x, y, 0) &= -\frac{16}{\pi^3} \left(\frac{dp}{dz} \right) \frac{(b/2)^2}{\mu} \\ &\times \sum_{n=1,3,\dots}^{\infty} \frac{(-1)^{(n-1)/2}}{n^3} \left(1 - \frac{\cosh(n\pi y/b)}{\cosh(n\pi a/2b)} \right) \cos\left(\frac{n\pi x}{b}\right) \\ v(x, y, 0) &= 0 \\ w(x, y, 0) &= 0 \end{aligned} \right\} \quad (2)$$

where the pressure gradient dp/dz is given in terms of the mean fluid velocity, u_m , by

$$u_m = -\frac{1}{3} \left(\frac{dp}{dz} \right) \frac{(b/2)^2}{\mu} \left[1 - \frac{192}{\pi^5} \left(\frac{b}{a} \right) \sum_{n=1,3,\dots}^{\infty} \frac{1}{n^5} \tanh\left(\frac{n\pi a}{2b}\right) \right] \quad (3)$$

An outflow boundary condition is specified at the channel outlet. A uniform heat flux is applied to all the external boundaries of the thin wall region while the liquid at the inlet is given a uniform temperature profile.

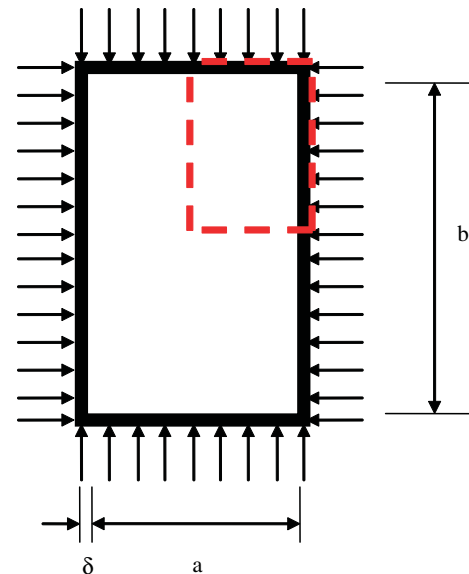


Fig. 1. Microchannel with thin conductive wall. The computational domain chosen from symmetry conditions is indicated with dashed lines.

Table 1
Dimensions of the microchannels investigated

α	a (μm)	b (μm)	D_h (μm)	L (mm)	Mesh (quarter domain)
1	200	200	200	120	$10 \times 10 \times 400$
2	200	400	267	120	$10 \times 20 \times 400$
3	200	600	300	120	$10 \times 30 \times 400$
4	200	800	320	120	$10 \times 40 \times 400$
5	200	1000	333	120	$10 \times 50 \times 400$
6	200	1200	343	120	$10 \times 60 \times 400$
7	200	1400	350	120	$10 \times 70 \times 400$
8	200	1600	356	120	$10 \times 80 \times 400$
9	200	1800	360	120	$10 \times 90 \times 400$
10	200	2000	364	120	$10 \times 100 \times 400$

Symmetry boundaries are used to reduce the extent of the computational model to a symmetric quarter-domain. There is no convective flux across a symmetry plane, so that the normal velocity component at the symmetry plane is thus zero. There is also no diffusion flux across the symmetry plane, rendering the normal gradients of all flow variables to be zero at the symmetry plane.

2.1. Solution method

The continuity equation and the Navier–Stokes equations in their steady, incompressible form, along with the associated boundary conditions were solved using the general-purpose finite-volume based computational fluid dynamics (CFD) software package, FLUENT [11]. An axially constant wall heat flux of 50 W/cm² with circumferentially constant wall temperature, i.e., the H1 thermal boundary condition, was applied on all four walls. This boundary condition has been shown by Lee et al. [3] to closely approximate the full three-dimensional conjugate heat transfer analysis with uniform heat flux applied to the bottom of the substrate as typically encountered in power electronics. Water enters the microchannels with a fully developed velocity profile at a temperature of 300 K. Only flow rates in the laminar regime were considered. The standard scheme was used for pressure discretization. The SIMPLE algorithm was employed for velocity–pressure coupling in the multi-grid solution procedure. The momentum and energy equations were solved with a first-order upwind scheme.

The entire domain was meshed using hexahedral elements. For the microchannel with aspect ratio of five, for example, a computational grid of 10 × 50 × 400 cells (for quarter channel) was used. The channel width and height were meshed with a uniform grid while the channel length was meshed with a double successive ratio of 1.02. The meshes used for the different aspect ratios are included in Table 1.

The local heat flux and local temperature distributions are obtained from the numerical simulations. With these quantities, the local convective heat transfer coefficient, $h(z)$, can be evaluated using the following equation:

$$h(z) = \frac{1}{A(z)} \frac{q(z)}{\sum_{x,y} [T_w(x,y,z) - T_m(z)] dA(x,y,z)} \quad (4)$$

where $A(z)$ and $q(z)$ are the total local heat transfer area and total local heat input, respectively, as defined below

$$A(z) = \sum_{x,y} dA(x,y,z) \quad (5)$$

$$q(z) = \sum_{x,y} q''(x,y,z) dA(x,y,z) \quad (6)$$

In Eq. (4), $T_w(x,y,z)$ is the local wall temperature and $T_m(z)$ is local fluid bulk-mean temperature given by

$$T_m(z) = T_{in} + \frac{1}{\dot{m} C_p} \sum_{x,y,z} q''(x,y,z) dA(x,y,z) \quad (7)$$

The local Nusselt number, $Nu(z)$, can then be calculated using

$$Nu(z) = \frac{D_h}{k_f} \times h(z) \quad (8)$$

The average Nusselt number, $Nu_{ave}(z)$, can similarly be computed as

$$Nu_{ave}(z) = \frac{D_h}{k_f} \times \frac{1}{\sum_{x,y,z} dA(x,y,z)} \times \frac{\sum_{x,y,z} q''(x,y,z) dA(x,y,z)}{\sum_{x,y,z} [T_w(x,y,z) - T_m(z)] dA(x,y,z)} \quad (9)$$

2.2. Grid-independence

The meshes tabulated in Table 1 for the different channel aspect ratios were verified to result in grid-independent results. As an example, local Nusselt numbers of 6.21, 6.15, and 6.11 were obtained for a channel aspect ratio of 5 with mesh sizes of 5 × 25 × 200, 10 × 50 × 400 and 15 × 75 × 600, respectively, at $z = 60$ mm when $Re = 1100$. The local Nusselt number changed by 1.7% from the first to the second mesh, and only by 0.7% upon further refinement to the finest grid. Hence the intermediate (10 × 50 × 400) grid was chosen. This grid-independence is also illustrated graphically in Fig. 2. Grid-independence was similarly established for the meshes selected for the other channel aspect ratios.

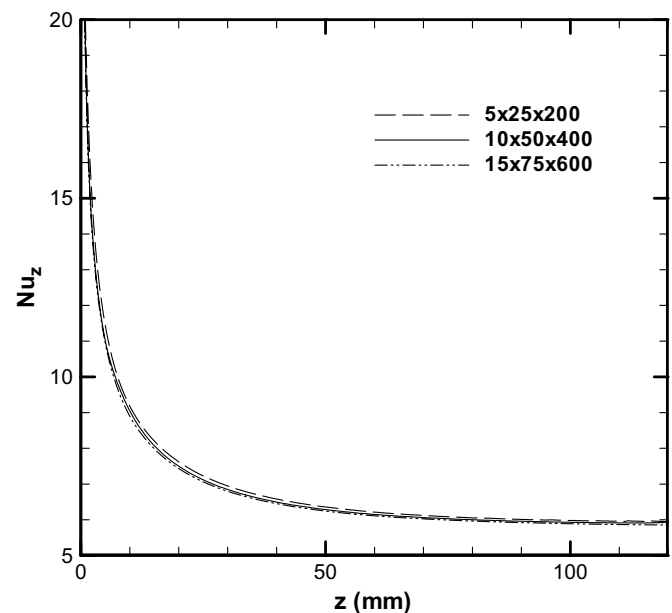


Fig. 2. Variation of local Nusselt number with axial distance for each of the different grids considered.

3. Results and discussion

3.1. Local and average Nusselt numbers

Fig. 3 shows the local Nusselt number as a function of dimensionless axial distance, $z^* = z/(Re Pr D_h)$ and aspect ratio (α). As expected due to the growth of the thermal boundary layer, the Nusselt number is very high at the beginning of the entrance region, but rapidly decreases and asymptotically approaches the fully developed values, given by the formula [8],

$$Nu_\infty = 8.235 \left(1 - \frac{2.0421}{\alpha} + \frac{3.0853}{\alpha^2} - \frac{2.4765}{\alpha^3} + \frac{1.0578}{\alpha^4} - \frac{0.1861}{\alpha^5} \right) \quad (10)$$

An interesting feature characteristic of rectangular channels is that the local heat-transfer conductance varies around the periphery and approaches zero at the square corners. This implies that the heat flux goes to zero at the corners [12]. Svino and Siegel [13] investigated the effect of unequal heat addition on adjacent sides of rectangular channels and found that poor convection due to low velocities in the corners and along the narrow wall causes peak temperatures to occur at the corners. Also, lower peak temperatures occur when only the longer sides are heated. This is reflected in the increase in local Nusselt number in the microchannel at a larger aspect ratio, since the relative importance of the narrow walls and corners diminishes with increasing aspect ratio.

The dimensionless thermal entrance length, z_{th}^* , defined as the distance required over which the local Nusselt number, Nu_z , drops to 1.05 times the fully developed value, Nu_∞ , can be determined from the results. Fig. 4 shows these values as a function of the channel aspect ratio. A larger aspect ratio channel is observed to have a shorter

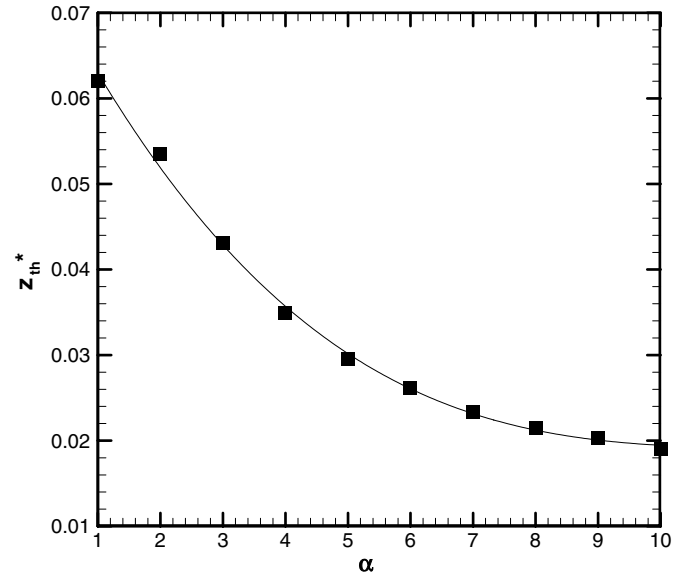


Fig. 4. Dimensionless thermal entrance length as a function of the channel aspect ratio. The curve-fit in the plot represents Eq. (11).

dimensionless thermal entrance length. The following relationship was curve-fit to the results with a correlation coefficient (R^2) of 1:

$$z_{th}^* = -1.275 \times 10^{-6} \alpha^6 + 4.709 \times 10^{-5} \alpha^5 - 6.902 \times 10^{-4} \alpha^4 + 5.014 \times 10^{-3} \alpha^3 - 1.769 \times 10^{-2} \alpha^2 + 1.845 \times 10^{-2} \alpha + 5.691 \times 10^{-2} \quad (11)$$

This correlation can be used to demarcate the developing regime from the fully developed regime. For accurate prediction of the heat transfer performance of a microchannel heat sink, the entrance effects need to be taken into account in the developing section, beyond which, the fully developed analysis is valid.

The average Nusselt number is shown in Fig. 5 as a function of dimensionless axial distance and aspect ratio. As in the case of the local Nusselt number, the average value starts high and decreases rapidly with downstream distance.

3.2. Nusselt number correlations

The following generalized correlation for local Nusselt number was obtained as a function of axial distance and channel aspect ratio from a regression analysis with a residual tolerance of 1×10^{-10} applied to 4000 computed values:

$$Nu_z = \frac{1}{C_1(z^*)^{C_2} + C_3} + C_4, \quad \text{for } 1 \leq \alpha \leq 10, \quad z^* < z_{th}^* \quad (12)$$

in which

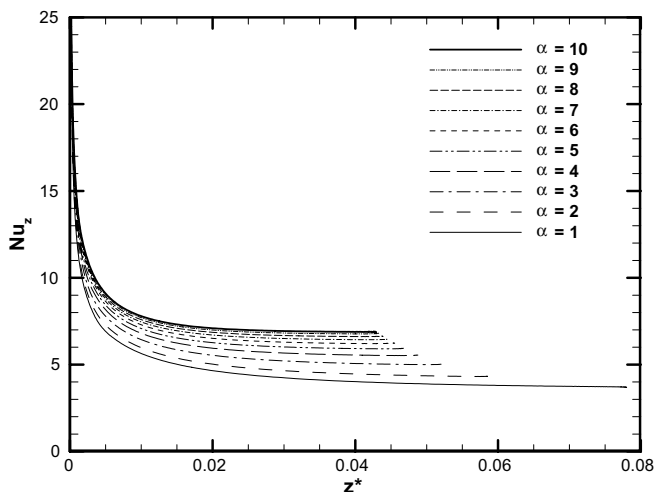


Fig. 3. Local Nusselt number as a function of dimensionless length and aspect ratio.

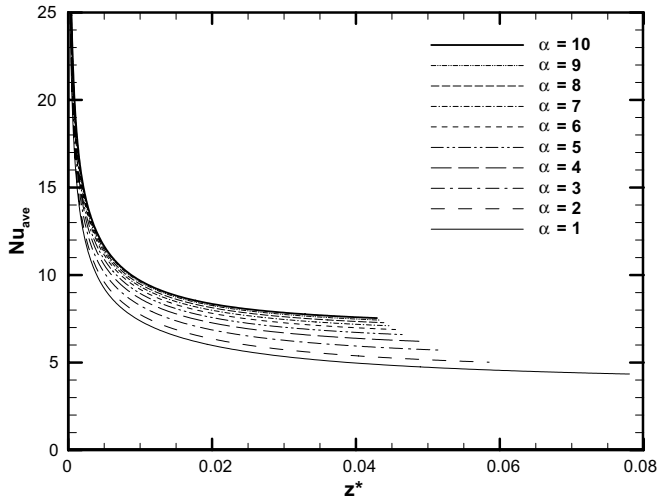


Fig. 5. Average Nusselt number as a function of dimensionless length and aspect ratio.

$$C_1 = -3.122 \times 10^{-3} \alpha^3 + 2.435 \times 10^{-2} \alpha^2 + 2.143 \times 10^{-1} \alpha + 7.325$$

$$C_2 = 6.412 \times 10^{-1}$$

$$C_3 = 1.589 \times 10^{-4} \alpha^2 - 2.603 \times 10^{-3} \alpha + 2.444 \times 10^{-2}, \text{ and}$$

$$C_4 = 7.148 - 1.328 \times 10^1 / \alpha + 1.515 \times 10^1 / \alpha^2 - 5.936 / \alpha^3$$

The fitted correlation has a very small average residual of 2.697×10^{-13} and an excellent correlation coefficient (R^2) of 0.999. This correlation is applicable when the flow is thermally developing, i.e., when $z^* < z_{th}^*$ from Eq. (11). Beyond z_{th}^* , the flow can be assumed to be fully developed, and Eq. (10) should be used instead.

A generalized correlation of very similar form was obtained for the average Nusselt number as well:

$$Nu_{ave} = \frac{1}{C_1 (x^*)^{C_2} + C_3} + C_4, \quad \text{for } 1 \leq \alpha \leq 10, \quad z^* < z_{th}^* \quad (13)$$

where

$$C_1 = -2.757 \times 10^{-3} \alpha^3 + 3.274 \times 10^{-2} \alpha^2 - 7.464 \times 10^{-5} \alpha + 4.476$$

$$C_2 = 6.391 \times 10^{-1}$$

$$C_3 = 1.604 \times 10^{-4} \alpha^2 - 2.622 \times 10^{-3} \alpha + 2.568 \times 10^{-2}, \text{ and}$$

$$C_4 = 7.301 - 1.311 \times 10^1 / \alpha + 1.519 \times 10^1 / \alpha^2 - 6.094 / \alpha^3$$

The fitted correlation again has a very small average residual of -9.684×10^{-13} and an excellent correlation coefficient (R^2) of 0.999.

3.3. Comparison with existing results for conventional channels

The numerical work of Wibulswas [5] perhaps most closely resembles the geometry (rectangular cross-section) and

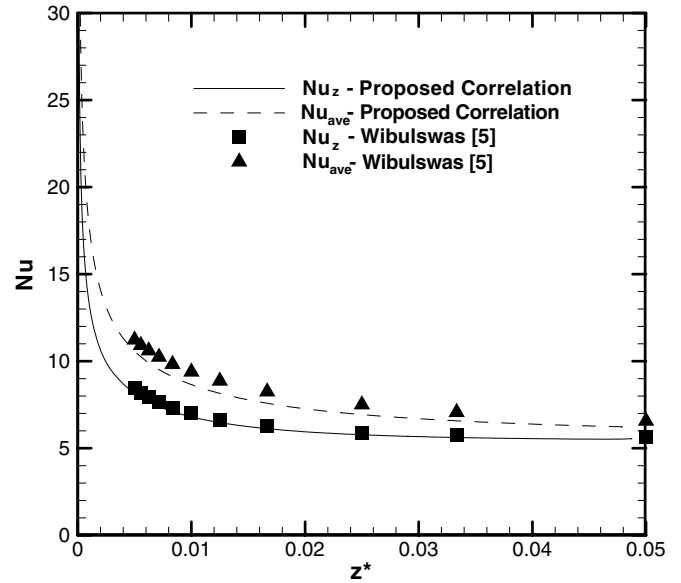


Fig. 6. Comparison of present work with that of Wibulswas [5] for $\alpha = 4$.

nature of the flow (thermally developing, with H1 boundary condition) considered in the present study. Predictions from the correlations proposed based on the results from the present study are compared to those of Wibulswas in Fig. 6, for a channel aspect ratio of four. While the agreement is, in general, quite good, the deviations between the two sets of results may be attributed to the very coarse mesh used in [5]. Also, Wibulswas reported very few data points near the channel entrance. In addition, the present work considers a much wider range of channel aspect ratios, and presents generalized correlations unlike the limited data available in [5].

Perkins et al. [14] proposed the following correlation based on their experimental measurements of local Nusselt number for a square duct:

$$Nu_z = [0.277 - 0.152 \exp(-38.6z^*)]^{-1} \quad (14)$$

The proposed correlations from the present study agree well with predictions from Eq. (14), as shown in Fig. 7, except in the region close to the channel inlet. Experimental data in this region were sparse, and therefore, the deviation in this region is not surprising.

Chandrupatla and Sastri [15] numerically analyzed the thermal entrance length problem for a square duct with the H1, H2 and T boundary conditions. While they used finer grids than those of Wibulswas [5], and their results agree closely with the proposed correlation as shown in Fig. 7, limited data were reported in the region close to the channel entrance. Lee et al. [3] deduced the dimensionless entrance lengths in past experimental studies on microchannel heat sinks and observed that the majority of these studies encounter thermally developing state ($0.003 \leq z^* \leq 0.056$). As such, resolving the heat transfer in the entrance region is very important for the accurate prediction of microchannel heat sink performance.

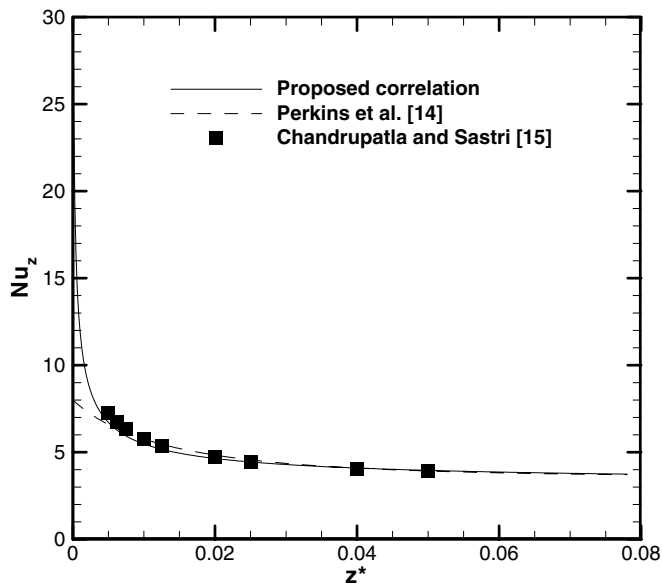


Fig. 7. Comparison of present work with that of Perkins et al. [14] and Chandrupatla and Sastri [15] for $\alpha = 1$.

It may be noted that the large aspect ratio ($\alpha > 4$) channels considered in this study are desirable due to the improvement in heat transfer coefficient with increasing aspect ratio. These deeper channels are readily fabricated by deep reactive ion etching (DRIE). Very limited information was available for the larger aspect ratios in the literature, and the correlations proposed here are important for the design and optimization of microchannel heat sinks.

3.4. Comparison with experimental data for microchannels

The proposed correlation for average Nusselt number was compared against experimental results obtained for microchannels in [3]. The experiments undertook a systematic investigation of single-phase laminar heat transfer in rectangular microchannels of widths ranging from 194 to 534 μm , with the channel depth being nominally five times the width. The test pieces were made of copper, and deionized water was used as the working fluid. In the experiments, only three walls were heated, with the top wall being made of an insulating material. The results from

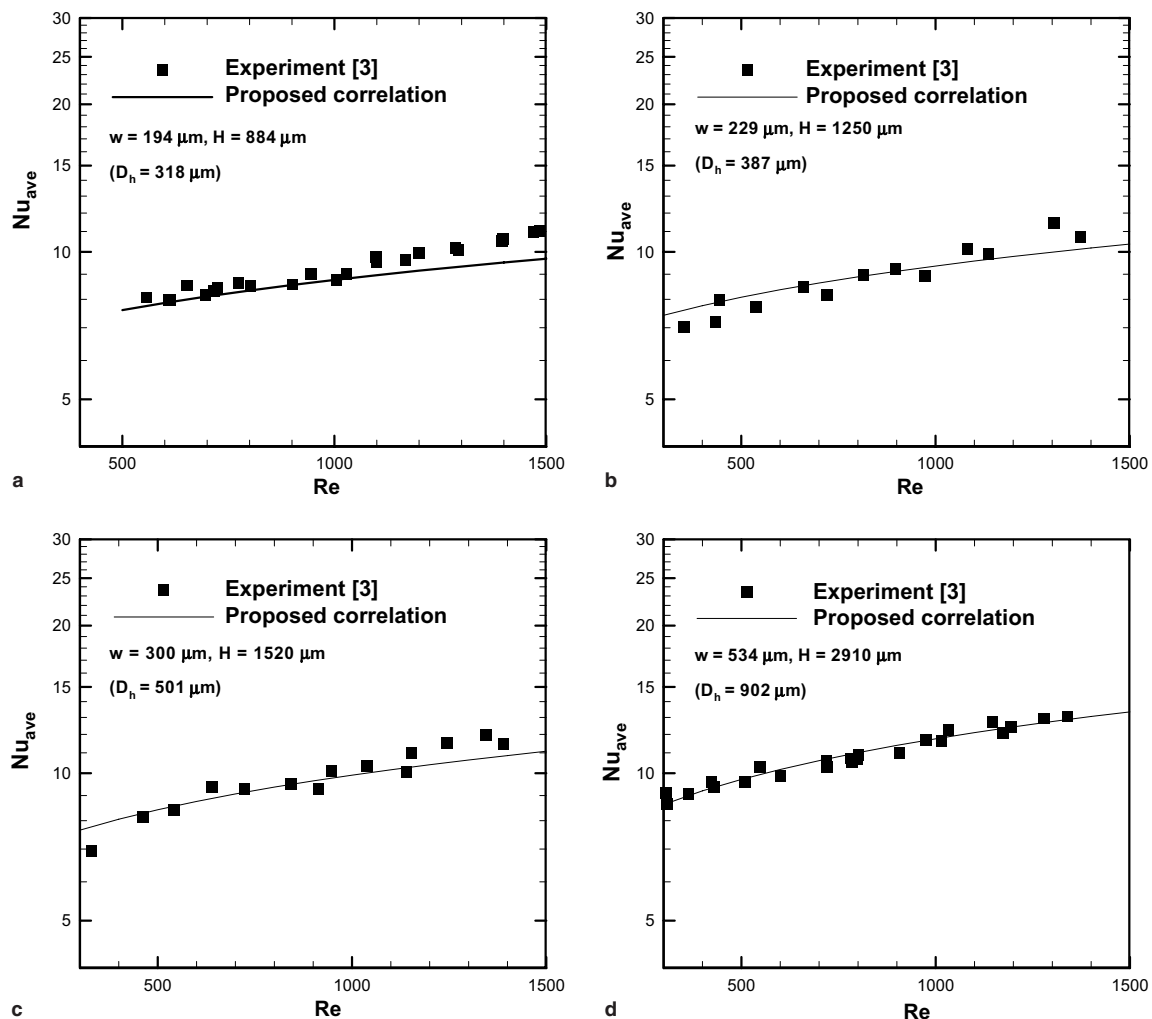


Fig. 8. Comparison of proposed correlation with experimental results of Lee et al. [3].

the present work, which considers heat transfer from all four sides of the rectangular channels, were adapted for comparison with the experiments using the correction factor proposed by Phillips [16]:

$$Nu_{z,3} = Nu_z \times (Nu_{\infty,3}/Nu_{\infty}) \quad (15)$$

This equation assumes that the ratio of developing Nusselt numbers for the three- and four-sided heating cases is identical to the value in fully developed flow. The fully developed Nusselt number for four-sided heating is given in Eq. (10) while that for the three-sided heating case can be approximated by the following formula [8]:

$$Nu_{\infty,3} = 8.235 \left(1 - \frac{1.883}{\alpha} + \frac{3.767}{\alpha^2} - \frac{5.814}{\alpha^3} + \frac{5.361}{\alpha^4} - \frac{2.0}{\alpha^5} \right) \quad (16)$$

Fig. 8 shows the comparison for four different sets of experiments with microchannels of different dimensions. The agreement is very satisfactory, suggesting that the proposed correlations based on a classical, continuum approach may be employed in predicting heat transfer coefficients in microchannel heat sinks.

The development of these generalized Nusselt number correlations could significantly aid in the design of microchannel heat sinks as this allows accurate predictions of their thermal performance without the need for full three-dimensional conjugate heat transfer analyses.

4. Conclusions

Local and average Nusselt numbers in the laminar flow of a Newtonian fluid through rectangular microchannels is investigated. Microchannel heat sinks are found to be best represented by the so-called H1 thermal boundary condition. Numerical simulations based on the finite volume method were conducted to predict steady, laminar heat transfer coefficients in hydrodynamically developed but thermally developing flow. Generalized correlations for both the local and average Nusselt numbers in the thermal entrance region are proposed. Predictions from these correlations compare very favorably with previous computational and experimental results for conventional channels, as well as with experimental results for microchannel heat

sinks. The proposed correlations allow accurate predictions of the thermal performance of microchannel heat sinks.

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